

Objectives

High level Objectives:-

- ⑧ Six Product Display bases that
- Detect product presence
 - lights base leds
 - Controls Video feed on TV Screen according to what product is lifted from product base

Technical Requirement

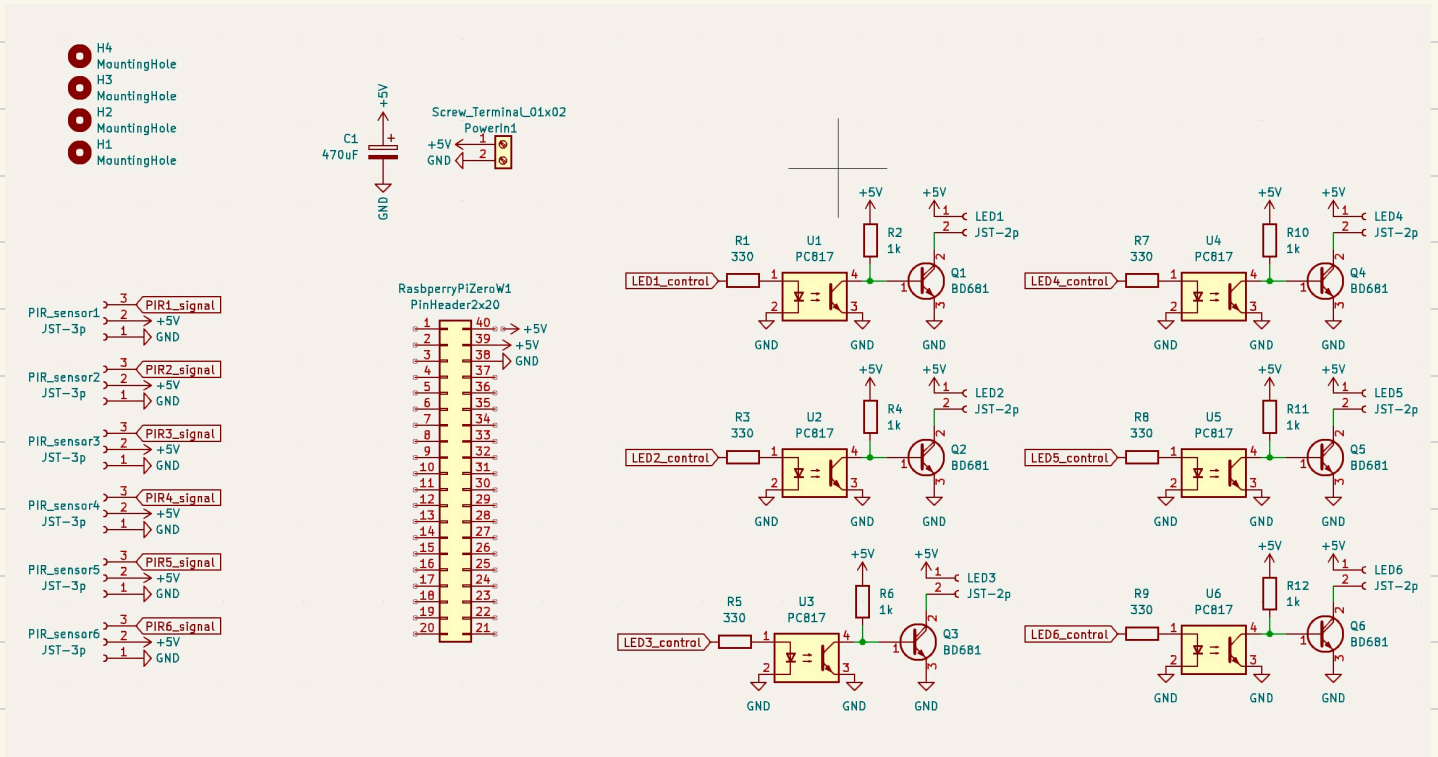
- ① Sensor to detect product presence
- ② Power Switching circuit to control product base led
- ③ Main Controller to
 - Read sensors (≈ 6 GPIOs)
 - Control leds (≈ 6 GPIOs)
 - View videos through HDMI

Proposed Solution

⊗ Hardware :

- 1- PIR Proximity sensor : to detect product presence
 - 2- LED Motion Sensor Wall light : Good looking Product base with built in led light (Operating @ 5v)
 - 3- IRFZ44n : MOSFET led power Switching
 - 4- Power supply (5v 2Amps) ^{leds} _{Math MCU}
 - 5- Raspberry Pi Zero W :
 - ↳ 40 GPIOs $\rightarrow$ 6 PIR proximity, 6 relay switching
 - ↳ Mini-HDMI connector
(will need to buy mini-HDMI male to HDMI male cable)
 - 6- PCB : for whole system
- ⊗ TV is provided by client
- ⊗ Shelf configuration by client

Initial Schematic



Software:

⊗ A Python Program to:

1- Control GPIOs for $\nearrow$ PIR proximity relay control

2- Output different Videos through HDMI

Action Steps

- 1- Figure out how to mount the PIR proximity sensors into the LED Motion Sensor Wall light. ← **Most risk!!**
- 2- Develop & test proposed solution on bread board. (including NMOS switching circuit)
- 3- Design & Manufacture PCB
- 4 - Buy & Assemble All hardware.
- 5 - Final testing

BOM

⊗ Minor changes
might apply

No.	Name	Cost	Quantity	Total Cost	Description	Store
1	PCB	200	1	200		
2	Raspberry Pi Zero W	2000	1	2000		
3	Pin Header 1x20	6	2	12		
4	mini-HDMI to HDMI cable	150	1	150		imedia store
5	Resistors	1	12	12	6x1k, 6x330	
6	Capacitor	5	1	5	470uF	
7	PC817	3	6	18	OptoCoupler	
8	BD681	4	6	24	Power Darlington NPN	
9	Terminal Block	3	1	3	Power In	
10	JST 3P Female	1	6	6		micro-ohm
11	JST 3P Male	6	6	36		micro-ohm
12	JST 2P Female	1	6	6		micro-ohm
13	JST 2P Male	6	6	36		micro-ohm
14	Cable 3-wire			0	for PIR sensor	
15	Cable 2-wire			0	for LED	
16	PIR proximity sensor	75	6	450		
17	LED Motion Sensor Wall Light	85	6	510		Amazon
18	Power Supply 5v 10Amp	390	1	390		micro-ohm
19	AC Wire	6	1	6		
20	AC Socket	20	1	20		
**	Delivery Fees	300	1	300		
<>	Design, Testing and Manufacturing	6000	1	6000		
	TOTAL			10184		

Gantt Chart

